AR vision (CSI)

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1. AR Vision
2. Preview of the Effect

1. AR Vision

AR (Augmented Reality) vision is a technology that overlays virtual information onto the real world.

An AR system has three prominent characteristics: ① the integration of information from the real world and the virtual world; ② real-time interactivity; ③ adding and positioning virtual objects in a three-dimensional space.

By using cameras, sensors and computer vision algorithms, an AR vision system can capture and analyse the real environment in real time and overlay the computer-generated images, text or other virtual content onto the real scene the user is looking at, thereby enhancing the perception and the interactive experience.

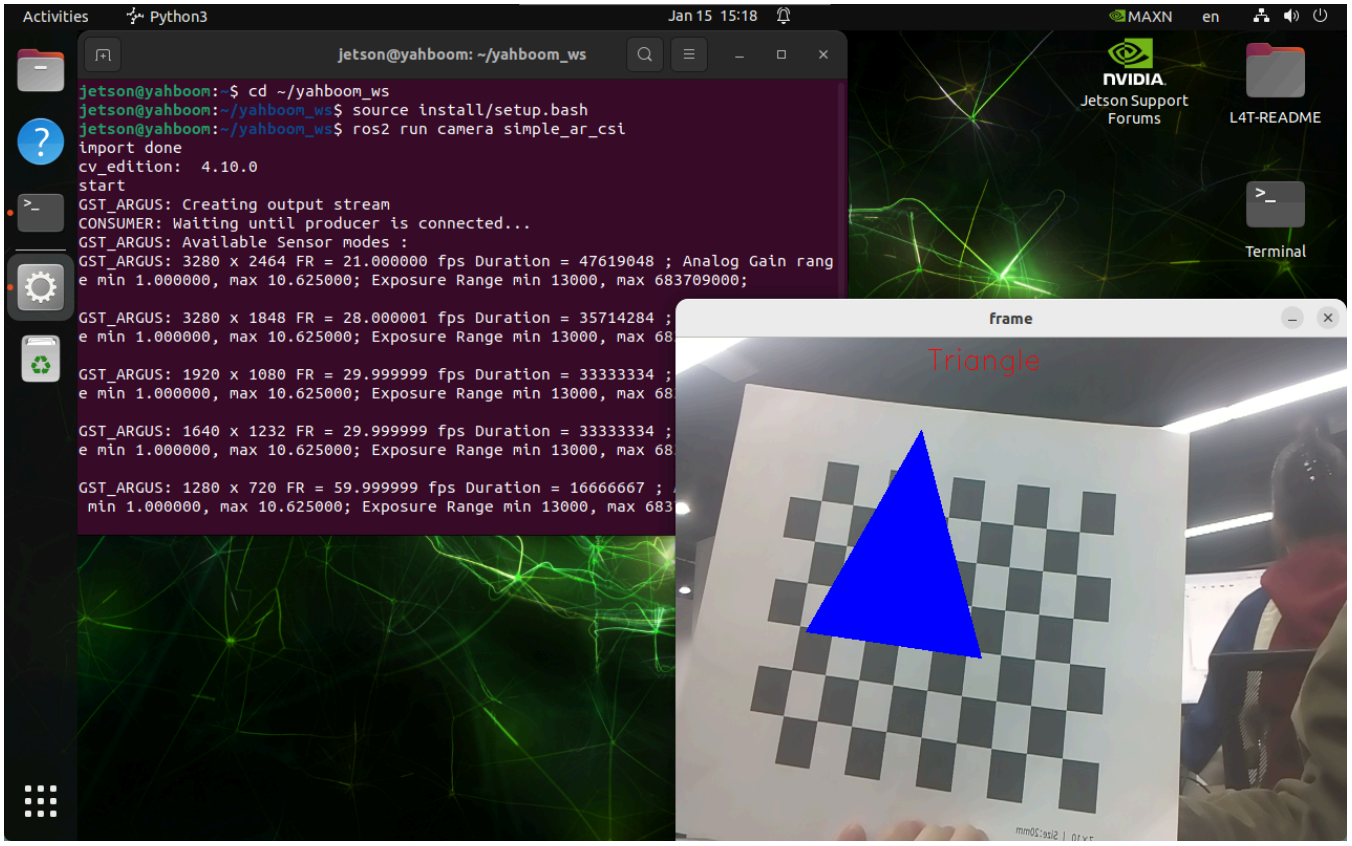
2. Preview of the Effect

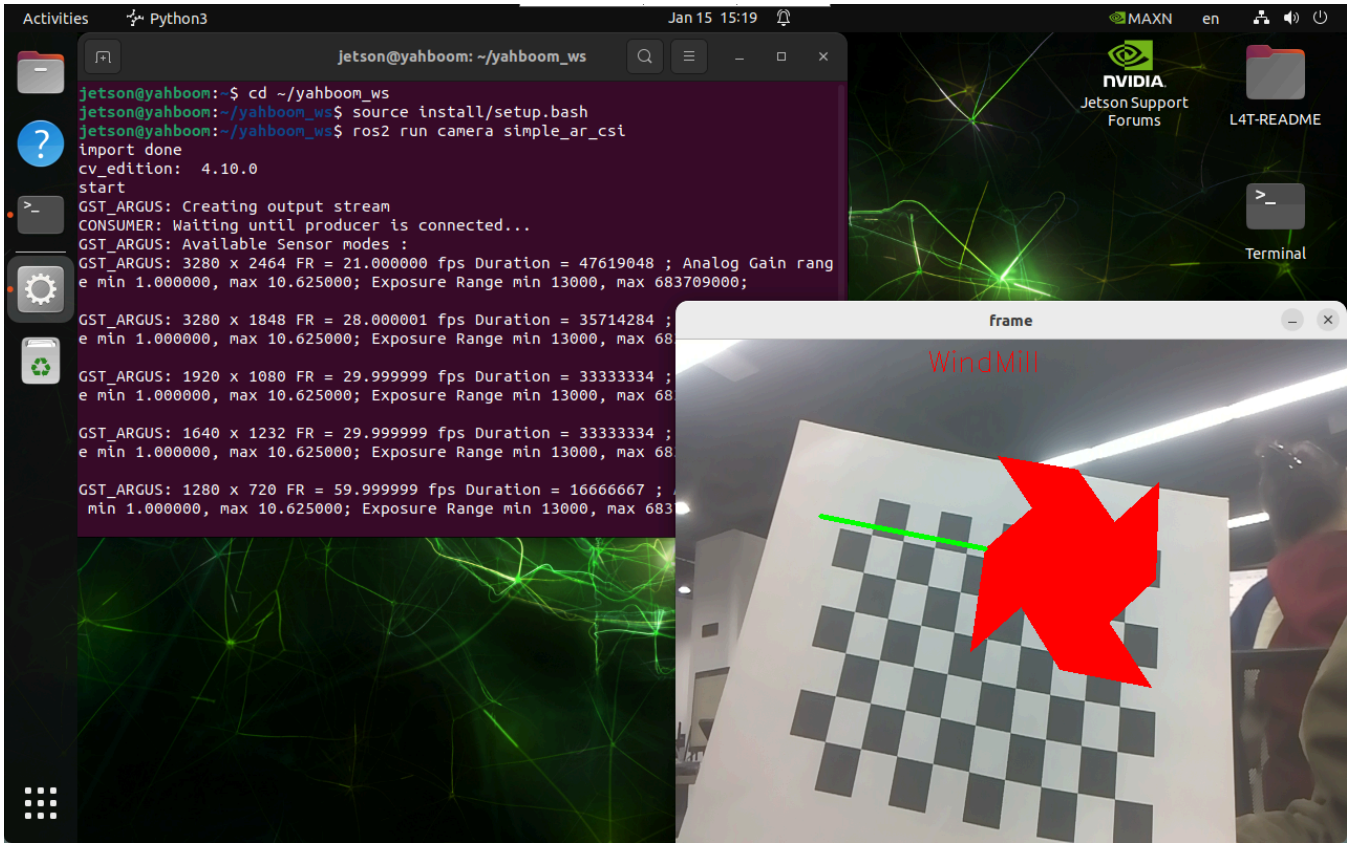
Click the preview window and press `f` to switch the virtual 3D object model.

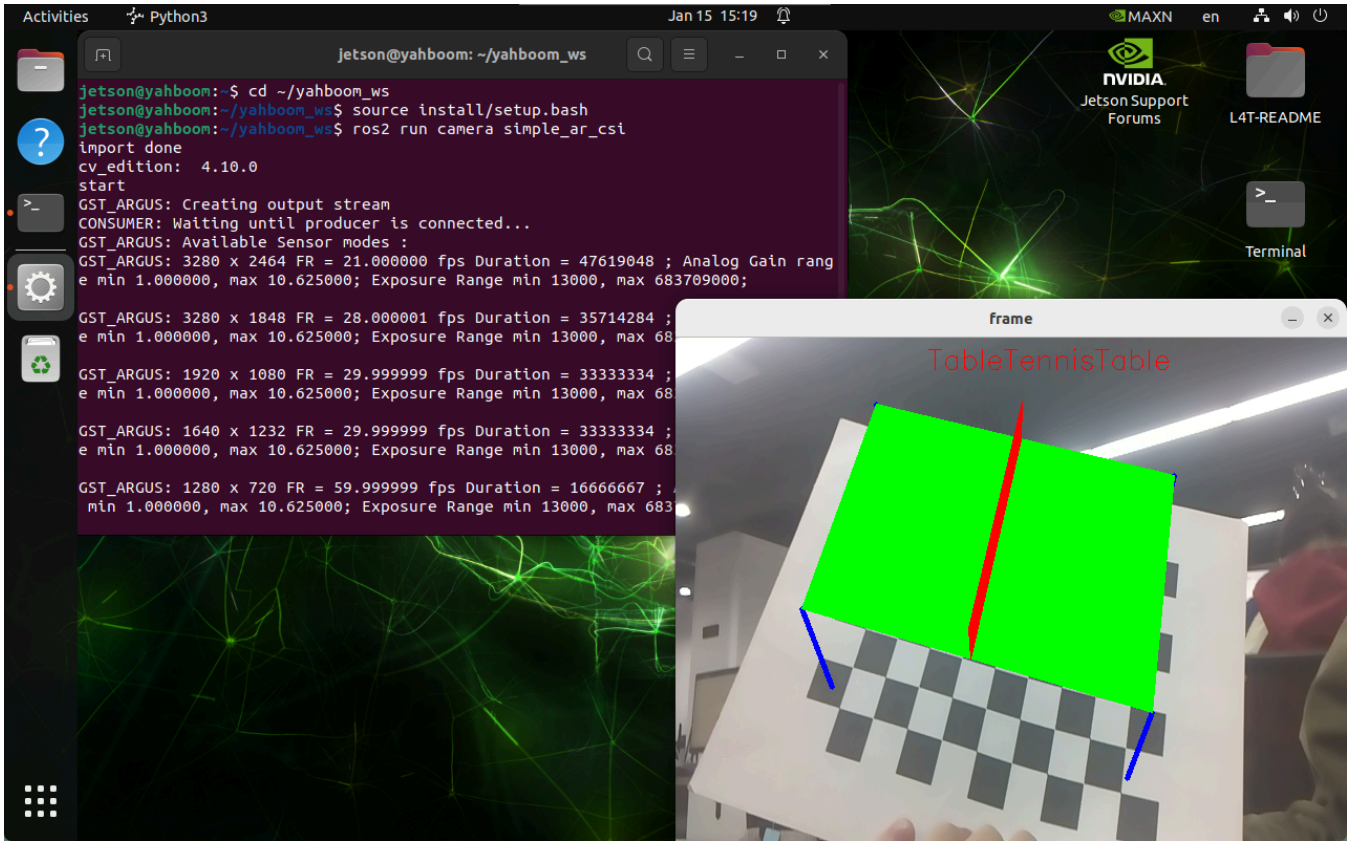
```
cd ~/yahboom_ws
```

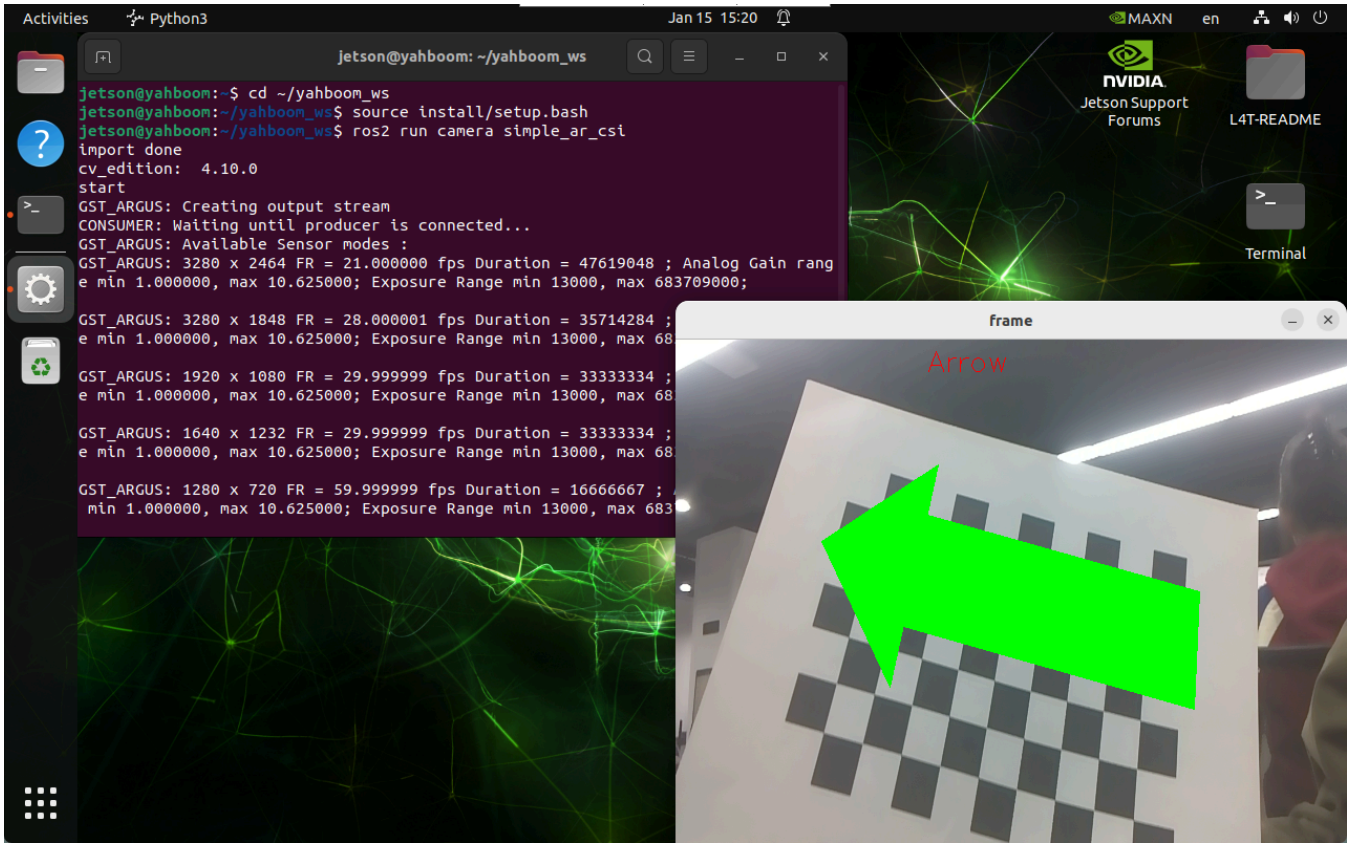
```
source install/setup.bash
```

```
ros2 run camera simple_ar_csi
```


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jetson@yahboom: ~/yahboom_ws
jetson@yahboom:~$ cd ~/yahboom_ws
jetson@yahboom:~/yahboom_ws$ source install/setup.bash
jetson@yahboom:~/yahboom_ws$ ros2 run camera simple_ar_csi
import done
cv_edition: 4.10.0
start
GST_ARGUS: Creating output stream
CONSUMER: Waiting until producer is connected...
GST_ARGUS: Available Sensor modes :
GST_ARGUS: 3280 x 2464 FR = 21.000000 fps Duration = 47619048 ; Analog Gain rang
e min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;
GST_ARGUS: 3280 x 1848 FR = 28.000001 fps Duration = 35714284 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 1920 x 1080 FR = 29.999999 fps Duration = 33333334 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 1640 x 1232 FR = 29.999999 fps Duration = 33333334 ;
e min 1.000000, max 10.625000; Exposure Range min 13000, max 68
GST_ARGUS: 1280 x 720 FR = 59.999999 fps Duration = 16666667 ;
min 1.000000, max 10.625000; Exposure Range min 13000, max 683
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jetson@yahboom:~$ cd ~/yahboom_ws
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jetson@yahboom: ~/yahboom_ws
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